



If you express it in the form of a negation, you can expect an effect similar to misinformation

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The primary goal of the presented research was to investigate how processing post-event information affects memory of details in an event viewed on video. We used two forms of post-event information: classic misinformation (changing or implanting new information into memory) and a new form that involves a correct explicit or implicit negation of the existence of an object in the video. We followed the three-step procedure used in studies on the misinformation effect, with a final memory test that was either immediate or delayed by a week and consisted in indicating which objects appeared in the video. We replicated the misinformation effect. More importantly, in the delayed test condition, both explicitly and implicitly negated objects were falsely recalled more often than unmentioned objects. These results indicate that it is possible to induce negation-related false memories; they also show that memory is impaired by negated post-event information or misleading post-event information.

Key words: explicit negation; eyewitness memory; false memory; implicit negation; memory distortion; misinformation effect; negation memory; negation-related false memories; post-event information; witness testimony.

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Introduction

‘Saw the burglar up close and personal - no balaclava in sight! Hope my eyewitness account can help catch the culprit. #NoBalaclava #CaughtInTheAct.’ Imagine that you are the person who yesterday witnessed a burglary and is now reading this post. Like the author of the tweet, you also remember that the perpetrator was not wearing a balaclava on his face. The results of our study show that if you were interviewed later and asked if the perpetrator was wearing a balaclava, you would be less likely to answer correctly than an eyewitness who had not read

about the balaclava: ‘No, he was not.’ Our results suggests that both explicitly and implicitly negating the presence of an object produces an effect similar to the misinformation effect.

Intensive research on memory distortions has been conducted since the 1970s, since when it has been shown that exposure to misinformation can have a considerable influence on our thoughts, memories and actions. Greenspan and Loftus (2021) pointed out that media coverage abounds with a lot of misinformation about a given disease and its effects during global pandemics such as COVID-19.

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The term ‘infodemic’, which refers to a media environment with abundant reliable and unreliable information, has even been coined (Greenspan & Loftus, 2021, see also World Health Organisation, WHO, 2020). As a consequence, retractions and corrections of the information that reaches people have become commonplace. Knowledge needs to be updated, but correcting misinformation is challenging (M. S. Chan et al., 2017; Lewandowsky et al., 2012), though not impossible (Walter & Tukachinsky, 2020). It has also been shown that false news on social media reaches more people and spreads faster than the truth (Vosoughi et al., 2018). Unfortunately, research shows that when faced with an imminent threat (in this case, an electric shock), people’s ability to label negated associations as false is impaired, while memories of ‘true’ associations remain intact (Newman et al., 2021).

Crucial aspects affected by misinformation and its denial or correction are part of the field of forensic psychology and, in particular, eyewitness testimony. Because of the reconstructive nature of memory (Bartlett, 1932), eyewitness testimony is susceptible to various types of distortion, which can occur during a criminal trial. Like physical evidence, such as fingerprints on a smartphone, bloodstains on a carpet or e-mails secured on the offender’s computer, eyewitness memory traces can be contaminated, lost or even fabricated (Wells & Loftus, 2003).

This legal context of memory malleability is within the area of interest of cognitive psychologists such as Elizabeth Loftus. In the procedure developed by her research team (Loftus et al., 1978), participants first viewed slides depicting a crime or accident, then were presented with information about the event peppered with misinformation, and finally their memory of the original event was tested. The results of their studies show that exposing people to misinformation can have a negative effect on their memory of past material. This phenomenon is referred to as *the*

misinformation effect (Loftus, 2005). The results of dozens of subsequent studies confirm that information that eyewitnesses are exposed to after an event or during a trial can distort their memory (see, e.g. Loftus, 2005, for a summary). These memory distortions can include both fine details of events (e.g. Dalton & Daneman, 2006; Heath & Erickson, 1998; Takarangi et al., 2006) and entire false events (e.g. Chrobak & Zaragoza, 2008; Lindsay et al., 2004; Pickrell et al., 2016; Wade et al., 2002). In the legal context, this means that suggestive questioning or suggestive lineup procedures can lead to false testimony by eyewitnesses. For example, recollections of events that never occurred may be confused with recollections of actual events (see Wells & Loftus, 2003).

Nonetheless, eyewitness testimony has been considered one of the most persuasive types of evidence in criminal trials. (e.g. Benton et al., 2006). The severity of the problem is illustrated by the startling data from the Innocence Project (2023). They show that a total of 375 wrongful convictions have been overturned by DNA testing in the United States since 1989. In nearly 69% of these cases, witness misidentification was identified as a contributing factor to the wrongful conviction, making it the leading cause of innocent convictions.

Negation

The results of previous research that was conducted in a wide range of research paradigms using a variety of experimental tasks have shown that negation processing is typically difficult, time-consuming and resource-demanding (e.g. Deutsch et al., 2006; Dudschig et al., 2019; Ferguson et al., 2008; Grant et al., 2004; Just & Carpenter, 1976; Kaup et al., 2006; Wason, 1959, but see also Nieuwland & Kuperberg, 2008; for review see Kaup & Dudschig, 2020). What is more, it is also not always effective and sometimes produces the opposite effect than intended (e.g.

Maciuszek, 2013; Wegner et al., 1998). The harmful effects of negation are perfectly illustrated by the results of research on the ‘white bear effect’. In this experiment, participants who actively tried not to think about a white bear were haunted by exactly this image (Wegner et al., 1987). However, negation processing appears to not only use (or fall back on) the cognitive control mechanisms that are typically reported during action control (de Vega et al., 2016; Dudschig & Kaup, 2018; Wirth et al., 2019).

The examples are research on the backfire effect, in which correction of misinformation inadvertently increases belief in the misinformation compared to the pre-correction or no-correction baseline (Swire-Thompson et al., 2020). There are two types of backfire effects: the worldview backfire effect and the familiarity backfire effect, both of which appear to be based on different psychological mechanisms. The worldview backfire effect occurs when a correction challenges a person’s belief system (Cook & Lewandowsky, 2012), and the familiarity backfire effect occurs when misinformation is repeated as part of a correction (Schwarz et al., 2007). This correction, in which misinformation is repeated to invalidate it, can increase the familiarity of the information, and thus unintentionally increase belief in the claim (Prike et al., 2023). Recent meta-analyses suggest that the backfire effect occurs rather rarely and, if so, only under specific conditions (Ecker et al., 2022; Swire-Thompson et al., 2020, 2022). For example, Autry and Duarte (2021) found that when participants were not initially exposed to novel misinformation and were then exposed to a correction, it had an adverse effect, resulting in even greater reliance on the misinformation than if they were not exposed to either the misinformation or the correction. Autry and Duarte (2021) explain this familiarity backfire effect by arguing that the negation effect is context-dependent—that is, if the presupposition being denied is known or can be easily processed, the negation is licensed and may

produce the negation effect. Remarkably, Autry and Duarte (2021) found a significant backfire effect only in their first experiment; moreover, Prike et al. (2023) failed to replicate this effect in their three experiments.

Negation-related false memories

Other general cognitive mechanisms utilized in non-linguistic cognition involved in negation processing is memory (e.g. Deutsch et al., 2006; Fiedler et al., 1996; Mayo et al., 2004, 2014). A fairly new trend in negation research was initiated by Maciuszek and colleagues (Maciuszek et al., 2019a, 2019b; Maciuszek & Polczyk, 2017), focusing on the discovery of the paradoxical effect of negation. This effect, called negation-related false memories (NRFM), consists in a false memory that the negated object was actually present or that the negated behavior actually occurred.

Researchers initially focused on the effect of explicit negation, which is constituted in English by words such as no, not, never, nobody and affixes such as unhappy, as well as explicitly negative quantifiers such as scarcely, hardly, few, seldom, little, only (Clark, 1980). In a series of three experiments, Maciuszek and Polczyk (2017) demonstrated and replicated the NRFM effect. In their study, the participants were familiarized with a description of a house (or activities performed by a car driver) in which some items (or activities) were mentioned, some were explicitly negated, and some were not mentioned (i.e. they did not appear in the description and served as control items in the memory test). The negation was contained in the sentences that comprised the description played back to the participants—for example, ‘There is a living room in the house, but there is no porch’. The final memory test occurred either 5 min after the presentation of the description or 1 week later. The test asked participants to mark for each item whether it was ‘Present’ (i.e. listed in the recording as being present in the home) or ‘Not present’ in the home. In the 5-min delay condition, there were no statistically

significant differences in recognition between negated and unmentioned items. In contrast, in the 1-week delay condition, participants recognized negated items as present more frequently than unmentioned items (they committed more false recognitions). This pattern of results was repeated in all three experiments. Furthermore, in the second and the third experiments, the difference in the number of false recognitions of negated items did not differ significantly from the number of correct recognitions of items indicated as present in the description.

Subsequent studies (Maciuszek et al., 2019a) compared the effect of explicit and implicit negation. The latter is constituted by pragmatic inferences and includes expressions like *forget*, *fail*, *doubt*, *deny* (Xiang et al., 2016). Similarly, to the previous study, participants listened to a recording in which some objects were mentioned as present, some were mentioned as absent, and some were not mentioned at all. In one study condition, the absence of objects was expressed using explicit negation, while in the second condition it was expressed using implicit negation. The memory test took place either 5 min or 1 week after participants heard the recording of the house description. The results indicate that explicit and implicit negation led to almost identical false recall of negated items. However, items not mentioned in the recording (i.e. controls—namely, items that were neither mentioned nor negated in the recording) were more often recognized as present by participants exposed to implicit negation than by those exposed to explicit negation. The researchers indicated that the mechanism that was possibly responsible for the false recall of negated items could be that participants remember the item itself but forget the context in which it was present (an affirmative or negative utterance), hence items would be recalled as present only because they were mentioned.

The latest research on NRFM compared the memory effects of implicit negation and explicit negation and tested whether

processing the former type of negation requires more cognitive effort (Maciuszek et al., 2019b). Three experiments were conducted using the same methodology as that in the authors' previous study (Maciuszek & Polczyk, 2017; Maciuszek et al., 2019a)—that is, by presenting to the participants source material consisting of a recorded description of a house in which the existence of certain objects was negated explicitly or implicitly (depending on the group) or was not mentioned; then the number of false alarms committed by the participants in the memory test was compared. In the first experiment, however, participants independently read the source material at their own pace, which was measured. No differences were demonstrated between explicit and implicit negation in the number of false alarms for negated and unmentioned items, answer time or time spent reading the description. In the next two experiments, the researchers manipulated the time spent reading the source material (in the second experiment, the exposure time was fixed, unlike in the first experiment, where participants read the description at their own pace) or the information about the upcoming memory test (in the third experiment, participants were warned about the test, unlike in the other studies). The results of these three studies were summarized as indicating that 'implicit and explicit negation have very similar if not identical effects on memory' (Maciuszek et al., 2019b, p. 301), and both need comparable cognitive effort because they require a similar amount of processing time.

In addition to the paradigm described above, in order to investigate whether a statement in the original material that negates the existence of an object (or action) can increase false memories of that object (or action), there are other strands of research investigating the effect of negation on memory. In the context of the present experiment, the strand identified as 'the effect on memory of correct denying' is important (see Maciuszek & Polczyk, 2017). It examines the effect on participants'

memory of their negation of the existence of certain objects or object properties in a subsequent memory test. The sequence of the study was as follows: participants were exposed to original material (e.g. a video showing the interior of the house) in which there was no negation; they then answered questions about the existence of certain objects (Fiedler et al., 1996) or object features (Mayo et al., 2014). Correctly answering some of the questions required the participant to negate the existence of an object or object feature. For example, in a study by Fiedler et al. (1996), participants were asked whether there was a teddy bear in the living room (which there really was not); in a study by Mayo et al. (2014), participants in one condition were asked whether a sculpture in a glass case was blue, and in the second condition they were asked whether it was red (it actually was red). In the study by Fiedler et al. (1996), study participants correctly negated the presence of objects that were not present; in Mayo et al.'s study, participants correctly answered 'no' or 'yes', respectively. This was followed by a final memory test in which participants were asked whether some items were in the original material. Fiedler et al. (1996) revealed that although participants correctly negated the presence of absent objects in the first memory test, they failed to do so in the second memory test. For example, correctly negating the presence of a teddy bear in the living room in the first memory test resulted in participants incorrectly indicating the presence of a teddy bear in the second, final memory test. In a series of four experiments performed by Mayo et al. (2014), it was found that correct negation of a feature of an object led to greater memory loss for that object than correct affirmation of that feature. That is, participants showed greater memory loss regarding the sculpture in the display case when they were in the 'no' condition than when they were in the 'yes' condition. Interestingly, the rate of memory loss in the negation condition was still greater than for items not followed by a question. In

conclusion, repetition in the form of a question increased recall, but the increase was smaller in the 'no' response condition.

The study conducted by Mayo et al. (2014) fits into research on the misinformation effect. They investigated whether information received after the event impairs an existing memory. On the surface, the research paradigms of Mayo et al. (2014) and Fiedler et al. (1996) were similar, but the research results produced seemingly contradictory findings. The study by Fiedler et al. (1996) shows that negation of a non-existent object can lead to false memory of that object. Simply put, they demonstrated that when propositions are processed in the form of 'no-XYZ', participants are prone to losing the 'no' and remembering the 'XYZ'. The researchers explain this by suggesting that thinking about an object and negating it during the first memory test leads to the creation of a transient representation of that object, which is activated during the final memory test and results in false memories. However, in the Mayo et al. (2014) study, the negation of an *attribute* of a present object might make participants forget the *object*. The mechanism of negation-induced forgetting remains unexplored, but Mayo et al. (2014) speculate that it may be related to the inhibitory character of the negation process: inhibition of the negated feature appears to spread to the object to which the feature is attributed. That is, during the first memory test, the participants presuppose the negated feature and thus generate a transient memory representation of the object with the incorrect feature; they then correctly negate it, which leads to the object representation being inhibited and its presence being forgotten (Mayo et al., 2014).

Overview of the present study

Using the classic three-step procedure for investigating the misinformation effect (Loftus et al., 1978), the aim of our study was to replicate the NRFM effect and compare its magnitude with that of the misinformation effect.

As in previous NRFM studies (Maciuszek et al., 2019a, 2019b), we concentrated on explicit and implicit negation ‘as a function of informing the absence of objects and, more specifically, in the context of memory recognition of implicitly and explicitly negated objects.’ (Maciuszek et al., 2019b, p. 294). In our study, explicit negation means that we informed about the absence of an object by means of a sentence in which the expression ‘not’ or ‘no’ occurred (see examples in the Materials section). In a forensic context, consider a description of the crime scene, which people become familiar with – for example, when reading a newspaper and are later interrogated. The description of the same situation can be expressed by explicit or implicit negation: *The case was difficult to unravel. There were no witnesses at the crime scene. The perpetrator left no fingerprints. There were no traces of the victim’s blood* (explicit negation). or *All witnesses were absent. If the perpetrator had left fingerprints or if traces of the victim’s blood had been found, the case would have been easier to unravel* (implicit negation).

A secondary goal of the study was to examine whether the typicality of details moderated the effect of post-event information presented in the form of explicit, implicit and misinformation on recognition. We considered that the typicality of details in post-event information is a factor that has important implications for the psychology of eyewitness testimony. As mentioned earlier, eyewitnesses are exposed to different information from different sources. On the one hand, if the post-event information contains atypical details, this may increase the likelihood of memory distortion because atypical details are better remembered than typical ones (this is called the ‘typicality effect,’ e.g. Davidson, 1994). This could lead to the atypical details being included in the memory of the original event. On the other hand, in research on the NRFM (Maciuszek & Polczyk, 2017, Exp. 1), the typicality effect occurred only for not-mentioned items. In the case of mentioned and negated

items, the differences between typical and non-typical items were not significant. However, Maciuszek and Polczyk’s (2017) research design consisted of two phases: familiarization with the source material and a recognition test. It also included one type of negation (explicit negation). Thus, our goal was to investigate the role of typicality in a three-stage paradigm in which post-event information takes the form of two types of negations and misinformation.

Our study was therefore part of a narrow line of research investigating whether post-event information (PEI) can affect memory accessibility (Ayers & Reder, 1998; Wright et al., 2001). Post-event information has so far taken the following forms (for a detailed description, see Mayo et al., 2014; Wright et al., 2001): (a) changing one piece of information to another (e.g. in the original material someone is playing a guitar, and in the post-event material someone is playing a piano; Loftus et al., 1978); (b) implanting information into memory (e.g. the original material does not show a guitarist, but a guitarist appears in the post-event material, e.g. Loftus & Pickrell, 1995); (c) omitting in the PEI the information shown in the original material (e.g. a person playing a guitar appears in the original material but is not mentioned in the post-event material, e.g. Loehr & Marche, 2006); (d) the post-event material consists of questions about items or item features from the original material (Mayo et al., 2014). A correct answer to a question requires a ‘no’ answer (for example, the original material shows an acoustic guitar, and the question is whether the guitar was electric). In our study, we tested whether PEI influences memory in the first and second ways, and we proposed testing yet another form of PEI influence– namely, providing in the form of negation true information that was not mentioned at all in the original material. Referring to the example with the guitarist and the guitar, in our experiment there would be no guitarist in the original material, and PEI would say that ‘There was no guitarist’

(explicit negation) or ‘If only there were a guitarist’ (implicit negation).

Our study is similar to the original NRFM study in the sense that it also tests whether a statement that negates the existence of an object can increase false memories of that object. However, it differs from the original NRFM study in that the statement is included in the post-event material. Like Maciuszek (Maciuszek & Polczyk, 2017; Maciuszek et al., 2019a, 2019b), as material we used a coherent story that contained negation. Unlike the methodology of the studies by Mayo et al. (2014) and Fiedler et al. (1996), where participants generated negation on their own, in our study the negation was administered to participants in PEI.

Hypotheses

In the present research, the following hypotheses were formulated:

Hypothesis 1: The misinformation effect will be replicated: the number of ‘Present’ answers for misleading items (i.e. those mentioned in PEI but not present in the original material), will be significantly higher than that for unmentioned (control) items. The occurrence of the misinformation effect has already been confirmed so many times (see Loftus, 2005) that it is even sometimes referred to as one of the most important discoveries of psychology (Zaragoza et al., 2007).

Hypothesis 2: After a 1-week interval, both explicit and implicit negation will result in false recall of negated details as being present in the original material: the number of ‘Present’ responses in the 1-week delay condition will be higher for explicitly negated (**Hypothesis 2a**) and implicitly negated (**Hypothesis 2b**) objects than for unmentioned (control) items.

The assumption that a time delay is needed for the NRFM effect to manifest itself requires a comment. In all the NRFM studies to date, the effect in the immediate recall condition

was non-significant but was revealed in the delayed memory test condition (Maciuszek & Polczyk, 2017; Maciuszek et al., 2019a, 2019b). This is probably because the memory of negated items during the immediate memory test is still strong, thus there is not yet room for NRFM effects. It is indicated that the integration of negation during comprehension is usually delayed and requires more attention (Kaup & Dudschig, 2020). Increased attention during PEI familiarization may lead to better recall of PEI and fewer errors in the immediate memory test. This beneficial effect of attention does not occur in the delayed memory test condition, thus resulting in the NRFM effect. Since explicit and implicit negation appear to have the same effect on memory (Maciuszek et al., 2019a, 2019b), we assume that the NRFM effect will also manifest itself after a time delay in the case of implicit negation.

Hypothesis 3: Both the NRFM and the misinformation effect will be stronger for typical details than for atypical details because of the tendency to fill memory gaps with items that conform to schemas and scripts and are therefore typical. Schemas can enhance memory formation (Alba & Hasher, 1983; Stangor & McMillan, 1992) and help in understanding and retrieving information. People use schemas to compensate for lost, uncoded or missing information (Luna & Seco, 2008). The information extracted from schemas tends to be typical information – that is, information that frequently appears in similar situations (e.g. Brewer & Treyens, 1981; García-Bajos & Migueles, 2003). Previous research has demonstrated that, on the one hand, during recollection there are more hits for typical information if it is true for the given situation (i.e. the content of the schema corresponds to reality); on the other hand, there are also more false alarms for highly typical information if it is false for the given situation (the content of the schema does not correspond to the actual situation; Bower et al., 1979; Brewer & Treyens, 1981; García-Bajos & Migueles, 2003; Lampinen et al., 2001; Nakamura et al.,

1985). The relationship between the typicality of details and the misinformation effect has been studied by, for example, Holst & Pezdek (1992); Luna & Seco (2008); Smith & Studebaker (1996, Exp. 1). The results of these studies indicate that the misinformation effect is greater for details with high typicality than for details with low typicality (Luna & Seco, 2008; Smith & Studebaker, 1996, Exp. 1). For example, in the Luna and Seco (2008) study, participants incorporated more high-typicality information into memory than low-typicality information, regardless of whether it was true or false (in other words, for high-typicality details, there were more of both hits and false alarms). Their study used implanted and altered PEI; the results indicate that there were no differences between altered and added details in terms of the number of false alarms (Luna & Seco, 2008).

Exposure to post-event information leads to the inclusion of elements in memories that were not present in the original event (Lindsay & Johnson, 1989; Loftus et al., 1978). In the context of misleading PEI (changing or adding elements), this is explained in such a way that exposure to PEI leads to a more liberal response criterion based on a sense of familiarity and, consequently, acceptance of the misinformation (Hekkanen & McEvoy, 2002; Luna & Migueles, 2009). The effects of negating PEI have not yet been studied, but we predict that participants in this case will also produce more false alarms for typical details (explicitly or implicitly negated) than for atypical details because they will remember the detail but not the context in which it occurred (explicit or implicit negation). This can lead to the same effect as presenting PEI in the form of misinformation: an increased tendency to respond 'present' for typical items.

In conclusion, compared to atypical items of each of these types, we hypothesize that there will be more false alarms for typical items: misleading (**Hypothesis 3a**), negated explicitly (**Hypothesis 3b**), negated implicitly (**Hypothesis 3c**), but also unmentioned

(control) (**Hypothesis 3d**). Since details with high typicality are part of the schema (Bartlett, 1932), when presented in a recognition memory test, participants will be able to retrieve them from memory, either because they will recall that memory or because they will extract them from the schema. Similarly, as in other studies, we assume that participants will tend to respond that they had seen typical objects before (Martín-Luengo et al., 2013).

Method

Prior to data collection, a positive opinion from the Jagiellonian University Institute of Psychology Research Ethics Committee in Cracow, Poland was obtained as written consent for participation from all persons in the study.

Participants

Two hundred participants recruited through social media advertisements took part in the study and were paid 25 PLN net for their participation. Data from 159 people (64.2% women) aged 18–72 years ($M = 26.45$, $SD = 10.31$) were included in the analysis. Data from 41 participants were excluded on the basis of post-experimental interviews, because these subjects had previously participated in memory distortion studies or guessed hypotheses.

Materials

- '1' *Original material*: A silent video clip depicting the theft of a laptop, lasting 1 min 32 s (a detailed description of the clip is included in [Supplementary materials](#)).
- '2' *Post-event material*: A story (193 words) describing the content of the original video. Eight different stories were prepared. Each story contained four misleading details, four explicitly negated details (information that the

Table 1. Examples of how misinformation and explicit and implicit negations were introduced.

Detail	Examples of sentences with:		
	Misinformation	Explicit negation	Implicit negation
Binder (typical)	The office could use a printer, but there is a binder.	There is a calendar in the office, but no binder.	If there were a binder, the papers from the desk could be organised
Motorcycle (typical)	The thief turns the corner of the building, where her motorcycle is parked	Her motorcycle, on which she can quickly escape, is not there	If her motorcycle had been standing there, she could have quickly escaped
Collar (atypical)	The employee's neck is adorned with a collar.	The employee's neck is not adorned with a collar.	The employee has no ornaments around her neck, perhaps she will wear a collar after work
Trolleybus (atypical)	On the street, a trolleybus can be seen moving.	The street is empty, not even a trolleybus is running.	The street wouldn't be so empty if a trolleybus were running.

object was not in the video, while it was actually not in the video), four implicitly negated details (their absence was mentioned in a certain context, and they were indeed not in the video). We created two versions of the post-event material: one contained rather typical objects (such as a printer, sandwich, necklace), while the other included rather atypical objects (such as a cassette, collar necklace, poncho). In each of the versions of the story, four details were not mentioned (used as a control in the memory test), explicitly negated, implicitly negated or misleading (see Table 1 with examples of how misinformation and explicit and implicit negations were introduced).

Full counterbalancing was used so that each of the items appeared in each of the four conditions: misleading, explicitly negated, implicitly negated and not

mentioned. In sum, to assure counterbalancing there were four versions of each of the two conditions (typical and atypical items). The stories were recorded and played to the participants, who also followed the text on a computer screen.

‘3’ *Memory test*: The memory test consisted of 20 items, 16 of which related to critical details (four misleading, four explicitly negated, four implicitly negated and four not mentioned), and four buffer details, which were actually present in the video. Separate memory tests were created for typical and atypical details. Depending on which version was assigned to a participant, a particular detail might have served for this participant as misleading, explicitly negated, implicitly negated or a control item (not mentioned). The participants’ task was to mark whether the given item was present or absent in the original material—that is, in the video.

Procedure

Participants were tested individually using a computerized procedure. At the beginning, they were familiarized with the original material (they watched the video). Then, they solved the filler tasks (these were questionnaires, used for the purpose of another study), then they listened to the recorded post-event material – that is, one of the eight stories. The text of the story was displayed on their computer screen so that they could follow it with their eyes. They then continued to complete their filler tasks. Later, depending on the condition, they either were dismissed (delayed memory test condition) or proceeded to complete the memory test (immediate memory test condition). According to the instructions, they had to indicate, on the basis of what they remembered from the video, whether a given object was present or absent in the video they had watched. The 20 objects were listed in alphabetical order. Afterwards, they were asked about previous participation in psychological studies and their predictions about the real purpose of the study, after which they were debriefed. Participants in the delayed memory test condition took the test a week later and then answered the same questions as participants in the immediate memory test condition, after which they were debriefed.

Experimental design

A $2 \times 2 \times 4$ design was applied, with two between-participants factors: delay (immediate vs. 1 week) and the version of the story (typical vs. atypical items). There was one within-participants factor: type of information (misleading vs. explicitly negated vs. implicitly negated vs. not mentioned). The dependent variable was the number of ‘Present’ answers in the memory test; for each of the four types of information, the possible range of answers was from 0 to 4. For all four types of information, the ‘Present’ response was incorrect (representing a false alarm).

Power analysis

The experimental design used in the present research included two between-subjects factors (with two levels each), one repeated-measures factor (four levels) and interactions between the grouping factors with the repeated measures. The analyses were performed for small, medium and large effects according to conventions suggested by Cohen (1988): $\eta^2_p = .01$, $.06$ and $.14$, which correspond to Cohen $f = 0.1$, 0.25 and 0.44 . The software G*Power was used for the analyses (Faul et al., 2007). Assuming 80% of the required power, the sample sizes for the three effect sizes were 138, 26 and 12. The final sample size used in the present research ($N = 159$) assures adequate power even for small effects.

Results

The descriptive statistics across all conditions are presented in Table 2 and Figure 1.

Using the proportion of false alarms for the 16 critical items in the recognition test (four misleading, four explicitly negated, four implicitly negated and four not mentioned, serving as control) – that is, ‘Present’ responses – we performed an analysis of variance (ANOVA): 4 (type of information: misleading, explicitly negated, implicitly negated, unmentioned) $\times$ 2 (typicality: typical, atypical) $\times$ 2 (delay: immediate, 1 week). We formulated all of our hypotheses a priori and referred to specific combinations of factor levels. Consequently, we adopted the strategy of planned comparisons: we report the results of the ANOVAs for the main effects and interactions; however, when verifying particular hypotheses, we rely on simple effects as these are best suited to a priori hypotheses (Keppel & Wickens, 2004; Rosenthal & Rosnow, 1985; Rosnow & Rosenthal, 1989). In addition to the frequentist approach, we also used the Bayesian approach in our analyses, using the method proposed by Masson (2011). To interpret the resulting Bayes factors, we used the classification proposed by van Doorn et al.

Table 2. Means and standard deviations for the ‘Present’ answers across experimental conditions.

Delay	Typicality	Misleading <i>M (SD)</i>	Explicitly negated <i>M (SD)</i>	Implicitly negated <i>M (SD)</i>	Control <i>M (SD)</i>	<i>N</i>
Immediate	Typical	2.13 (1.03)	1.23 (0.96)	0.85 (0.99)	1.67 (0.96)	39
	Atypical	1.43 (1.26)	0.89 (0.94)	0.62 (0.79)	0.70 (0.91)	37
	Total	1.79 (1.19)	1.07 (0.96)	0.74 (0.90)	1.20 (1.05)	76
One week	Typical	1.98 (1.02)	1.98 (1.02)	1.91 (1.10)	1.59 (1.11)	44
	Atypical	1.72 (1.12)	1.31 (1.17)	1.31 (1.24)	0.92 (0.96)	39
	Total	1.86 (1.07)	1.66 (1.14)	1.63 (1.20)	1.28 (1.09)	83
Total	Typical	2.05 (1.02)	1.63 (1.06)	1.41 (1.17)	1.63 (1.03)	83
	Atypical	1.58 (1.19)	1.11 (1.08)	0.97 (1.10)	0.82 (0.93)	76
	Total	1.82 (1.13)	1.38 (1.09)	1.20 (1.15)	1.24 (1.06)	159

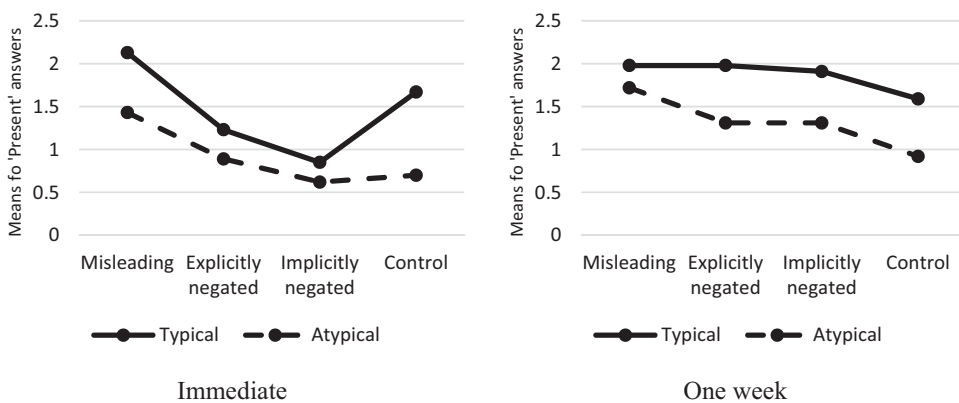


Figure 1. Mean numbers of ‘Present’ answers across experimental conditions.

(2021). According to this, a Bayes factor (BF) of less than 0.1 indicates strong evidence; a factor between 0.1 and 0.3 indicates moderate evidence for the null hypothesis (H0); a factor between 0.3 and 3 indicates weak/inconclusive evidence; a factor between 3 and 10 indicates moderate evidence; while a factor greater than 10 indicates strong evidence in favour of the alternative hypothesis (H1; van Doorn et al., 2021).

Overall, the type of information (misleading, explicitly negated, implicitly negated and unmentioned) significantly influenced the number of ‘Present’ answers, $F(3, 465) = 14.53, p < .001, \eta^2_p = .09, BF_{10} > 100$; it was higher in the case of misleading items than in the remaining three cases.

There were more false alarms for the misleading details ($M = 1.82$) than for unmentioned (control) items ($M = 1.24$), $F(1, 155) = 27.06, p < .001, \eta^2_p = .15, BF_{10} > 100$, indicating the detrimental effect of misleading post-event information on memory. This effect was present in both the immediate condition and the delayed condition: questions regarding misleading items in the description were more often answered as ‘Present’ than those regarding unmentioned items: $F(1, 155) = 13.05, p < .001, \eta^2_p = .08, BF_{10} = 49.25$; $F(1, 155) = 13.99, p < .001, \eta^2_p = .08, BF_{10} = 76.10$, respectively. There was no difference between the number of ‘present’ responses for misleading details between the immediate and delayed test conditions, $F(1, 155) = 0.14, p = .703$,

$\eta^2_p < .01$; also, the Bayesian analysis provided support for no between-condition differences ($BF_{10} = 0.09$). Significantly more false alarms for misleading items than for unmentioned items were committed for both typical and atypical details: $F(1, 155) = 7.20, p = .008, \eta^2_p = .04, BF_{10} = 2.94$ (the Bayes factor was rather moderate in this case); $F(1, 155) = 21.33, p < .001, \eta^2_p = .12, BF_{10} > 100$, respectively. Thus, these results support Hypothesis 1, which assumes the existence of the misinformation effect.

The interaction between the type of information (misleading, explicitly negated, implicitly negated and unmentioned) and delay was statistically significant, although the Bayesian analysis did not confirm it, $F(3, 465) = 6.71, p < .001, \eta^2_p = .04, BF_{10} = 0.08$. The planned comparisons revealed that, in the immediate condition, the items explicitly negated in the description were not more often indicated as 'Present' than were unmentioned items, $F(1, 155) = 0.68, p = .410, \eta^2_p = .00, BF_{10} = 0.11$. In contrast, the items implicitly negated in the description were significantly less frequently indicated as 'Present' than unmentioned items, $F(1, 155) = 8.69, p = .004, \eta^2_p = .05, BF_{10} = 6.06$. The difference between 'Present' indications for explicitly negated and implicitly negated items in the immediate condition was significant, $F(1, 155) = 5.16, p = .024, \eta^2_p = .03, BF_{10} = 1.09$. The explicitly negated items were more often ($M = 1.07$) indicated as 'Present' than implicitly negated ones ($M = 0.74$). The number of 'Present' indications for the unmentioned items did not differ significantly between the immediate and the 1-week delay conditions, $F(1, 155) = 0.21, p = .646, \eta^2_p = .00, BF_{10} = 0.09$.

The hypothesized NRFM effect was present in the delayed condition: items that were both explicitly negated ($M = 1.66$) and implicitly negated ($M = 1.63$) in the description were more often reported as 'Present' than items that were not mentioned at all ($M = 1.28$): $F(1, 155) = 7.25, p = .008, \eta^2_p = .04, BF_{10} = 3.00$; $F(1, 155) = 5.70, p = .018,$

$\eta^2_p = .04, BF_{10} = 1.44$, respectively. This pattern of results confirms that negated items tend to be remembered as present more often than unmentioned items, which supports Hypotheses 2a and 2b regarding the NRFM effect. From the perspective of frequency statistics, the results are statistically significant and support Hypotheses 2a and 2b regarding the NRFM effect. From the perspective of Bayesian statistics, the results remain inconclusive. The difference between explicitly negated items and implicitly negated items in the delayed condition was not significant, $F(1, 155) = 0.06, p = .805, \eta^2_p < .01$. Bayesian analysis supported the nonexistence of a difference, $BF_{10} = 0.08$.

Furthermore, in the delayed memory test condition, there were no statistically significant differences between the number of false alarms for misleading (that is, mentioned; $M = 1.86$) versus explicitly negated details, $F(1, 155) = 1.61, p = .206, \eta^2_p = .01$, as well as between the number of false alarms for misleading versus implicitly negated details, $F(1, 155) = 2.41, p = .123, \eta^2_p = .02$. The Bayesian analysis revealed that the Bayes factor for the differences between misleading and explicitly negated details remembered as 'present' was $BF_{10} = 0.18$, which means that there were no differences.

Similarly, when comparing misleading and implicitly negated details, the Bayesian analysis indicated that there were no differences ($BF_{10} = 0.27$).

In contrast to the significant interaction of delay with the type of information, the interaction of type of information with typicality was not significant, $F(3, 465) = 1.36, p = .253, \eta^2_p = .01, BF_{10} = 0.16$. There was a significant main effect of typicality on the number of 'Present' responses overall, $F(1, 155) = 31.20, p < .001, \eta^2_p = .17, BF_{10} > 100$. In general, typical items ($M = 1.67$) were reported as 'Present' more often than atypical ones ($M = 1.11$). For every type of information, there were more false alarms for typical

details than for their atypical counterparts. Thus, there were significantly more false alarms for misleading typical items than for misleading atypical items, $F(1, 155) = 7.38$, $p = .007$, $\eta^2_p = .45$, $BF_{10} = 3.16$, which supported Hypothesis 3a. More false alarms were noted for explicitly negated typical items than for explicitly negated atypical items, $F(1, 155) = 9.56$, $p = .002$, $\eta^2_p = .06$, $BF_{10} = 9.09$, which means Hypothesis 3b was supported. Likewise, there were more false alarms for implicitly negated typical items than for implicitly negated atypical items, $F(1, 155) = 6.19$, $p = .014$, $\eta^2_p = .04$, $BF_{10} = 1.76$, which supported Hypothesis 3c, but the Bayes factor did not support it. Finally, there were also more false alarms for unmentioned typical items than for unmentioned atypical items, $F(1, 155) = 27.01$, $p < .001$, $\eta^2_p = .15$, $BF_{10} > 100$, which means Hypothesis 3d was confirmed. Taken together, Hypothesis 3 was supported, according to which there should be more false alarms for typical misleading, explicitly negated, implicitly negated and unmentioned items than for atypical items of each of these types.

Finally, the three-way interaction of type of information, delay and typicality was not significant, $F(3, 465) = 1.87$, $p = .135$, $\eta^2_p = .01$, $BF = 2.30$.

Discussion

The goal of this study was to examine how memory concerning the presence of given items is affected by PEI in the form of misinformation or information that correctly negates, explicitly or implicitly, some items. The results of our study support the widely held view in the literature that ‘human memory is not a recording device, but rather a process of (re)construction that is vulnerable to both internal and external influences’ (Van Damme & Smets, 2014, p. 310). There have been hundreds of studies on misinformation over the past 40 years, most of which have focused on post-event information that implants new

memories or alters existing ones (Loftus, 2005), thus demonstrating that suggestion is a powerful tool that can elicit vivid false memories. Surprisingly little research, however, has addressed the question of whether negation can distort memories of witnessed events. To answer this question, in our experiment we applied a three-step procedure that is commonly used to study the misinformation effect. In the first phase, participants witnessed an event by watching a short video. In the second phase, they were exposed to misleading, explicitly negated and implicitly negated information about some critical details of the event (some details were also not mentioned and served as control items). This information was presented by means of a narrative. In the third and final phase, a recognition memory test about the original event was administered: participants were asked to indicate whether a given item was present in the video.

Our first hypothesis regarding replication of the misinformation effect was supported. Regardless of time delay and the typicality of details, participants were more likely to indicate that misleading items were present in the original watched material – that is, the video – than unmentioned items. Like numerous previous studies, our experiment demonstrated that people often remember misleading information from post-event information as part of the original witnessed event; that is, they are prone to incorporating external suggestions into their memory of the event (e.g. Belli et al., 1994; Calvillo et al., 2016; Loftus, 1979; Loftus et al., 1978; McCloskey & Zaragoza, 1985; Takarangi et al., 2006). According to the effect size classification proposed by Cohen (1969; see also Richardson, 2011), an effect for which the partial eta squared reaches a value of .1279 is considered large. Thus, the misinformation effect from our study, with a value of .15, can be interpreted as large.

The most important findings of the present experiment concern the NRFM effect. In the 1-week time delay condition, explicitly negated details were more often indicated as

present in the original material than unmentioned details. Similarly, implicitly negated details were more frequently indicated as present in the video after 1 week than unmentioned details. It should obviously be noted that these results should be treated with caution because the Bayes factor was not large. According to the adopted interpretation framework, this means that, in the case of explicitly negated details, evidence was weak at best in favour of the alternative hypothesis, which states that the number of 'Present' responses in the 1-week delay condition will be higher than for unmentioned items. In the case of implicitly negated details, the Bayes factor pointed to weak evidence in favour of the alternative hypothesis, which states that the number of 'Present' responses in the one-week delay condition will be higher for these objects than for unmentioned items.

In line with the Bayesian analyses, there is strong support for the idea that implicit and explicit negation have analogous effects on memory: there were no differences between implicit and explicit negation as regards the number of false alarms. As in the study by Maciuszek et al. (2019b), our results also suggest that these two types of negation can result in the same memory quality.

This result seems to be unexpected because most studies show that negation is more difficult to process, and when comparing explicit and implicit negation, the former is more difficult. One proposed explanation for the 'negation effect' assumes that the use of negation can lead to processing challenges, especially in early childhood, because of its negative connotation, which is strongly associated with prohibitions (Kaup & Dudschig, 2020). According to Kaup and Dudschig (2020), in the case of implicit negation, the negation marker responsible for unpleasant connotations with prohibitions is not present in the sentence; therefore, implicit negation leads to smaller negation effects than explicit negation. This hypothesis is supported by the results of several studies (Eifermann, 1961,

Jones, 1968; Wason & Jones, 1963, but see: Clark, 1974). This was also true for the immediate memory test condition in our study. If we consider the number of false alarms as an indicator of negation processing efficiency (the lower the number of false alarms, the better the negation processing), then in the immediate condition implicit negation indeed generates fewer errors than explicit negation. However, no such effect was observed for the delayed memory test condition, in which there was an NRFM effect. An explanation for why, in the delayed condition, both negations have the same effect on memory and on misinformation might be that participants forget the context in which a particular word appeared. It may be that, after a week, participants remember that the object appeared somewhere, but they do not remember the content of the original and subsequent material well enough (or they confuse the two sources) to be able to correctly reconstruct the source and context in which the object was mentioned. In subsequent studies, it would be worthwhile to test the context hypothesis – for example, using a memory test that employs cued recall or reminding subjects of the context in which the object appeared.

In our study, we used a set of typical and atypical details. As predicted, typical details were more often remembered as 'present' than atypical details. This relationship was significant for each type of detail: misleading, explicitly negated, implicitly negated and unmentioned. This is consistent with our prediction, based on findings from previous studies, that typical details may be more easily implemented into memory and that when memory gaps occur, the person may resort to schemata to fill them (e.g. García-Bajos & Migueles, 2003; Lampinen et al., 2001; Luna & Seco, 2008).

Our results can also be considered from the point of view of the violation of communication norms. The negations we used in our post-event materials negated side details. Participants may have felt that they were not

paying attention to the presence or absence of these details in the original material (because they were focused on the action of the video, for example). Denying the presence of these elements may be considered as unlicensed negation, which contradicts Grice's (1975) maxims of relevance in communication. Therefore, they could have perceived the negation of, for example, the presence of the mirror in the travel agency as unexpected or odd. As a result, participants may conclude that the negation of information that is irrelevant from their point of view has a hidden purpose, which makes them sceptical of the information, and although they may not remember whether the mirror was present or not, they begin to believe that it was. This may be especially true for details that fit a schema, such as a typical office or the clothing of a motorcyclist. However, in the absence of a pilot test on typical and atypical details, the conclusions we can draw in this regard are limited.

Practical implications, limitations and future directions

It is not hard to imagine that the situation described in the introduction really happened. Thus, the results of our study may have serious applied implications as they show the danger that lurks during interrogations: a witness exposed to PEI that implants or alters certain details about an event may inadvertently give false testimony, thus risking a potentially innocent person spending years in prison for something they did not do. Worse yet, an honest witness who encounters truthful PEI that something was not present or that something did not happen may later develop false memories about the presence of that thing or the occurrence of the event.

An undeniable weakness of the study was that we did not conduct a pilot study to confirm the selection of details considered typical and atypical. However, as mentioned previously, the analyses related to the effect of typicality of details on memory were an additional objective of the study.

As the typicality was manipulated as a between-participants design, one of the groups of participants had all critical statements with low typicality. This might have created some suspicions which in turn might have impacted the credibility of the post-event information. This must be regarded as a weakness of the present study.

From both a theoretical and a practical point of view, it seems important to clarify the boundary conditions for the generation of the NRFM effect. For example, future studies could use materials other than the video and audio recording used in the current study to examine the influence of the nature of the original material and critical items on the formation of memory distortions. In addition, it would seem interesting to compare the influence of PEI in the form of explicit and implicit negation against PEI in the form of questions. We are not thinking of questions with false presuppositions but about the form of sentences in PEI. For example, if an explicit negated PEI reads 'There was no kitten there', a question form of PEI could read 'Was there a kitten?'. There is an experiment in which PEI in the form of a question eliminated the misinformation effect, but in this study the final memory test was preceded by a warning (Lee & Chen, 2013). From there, ways to prevent NRFM are certainly worth exploring. Several methods have been tested to reduce the negative impact of misleading PEI on memory. These include warning (for review: Blank & Launay, 2014), change-detection (Bailey et al., 2021), Cognitive Interview (CI; e.g. Holliday et al., 2012), mental warm-up (Szpitalak & Polczyk, 2014) and reinforced self-affirmation (RSA; Szpitalak, 2022; Szpitalak & Polczyk, 2013). It is possible that some of these could prove effective in reducing the deleterious effect of negation on witnesses' memory.

This study used a three-step procedure to study the misinformation effect. Its shortcoming is that it does not allow us to check whether, at the time of the final memory test, the participant had both correct pieces of

information (from the original material and PEI) in their memory, or whether they remembered only one of them, had incorrect memories or had none at all. It was found that there are people who are aware of the discrepancies between original material and PEI but still succumb to the misinformation effect (Blank, 1998; Polczyk, 2017). It would therefore be worthwhile to design a procedure to distinguish such individuals in the context of NRFM. This is important because techniques that reduce the deleterious effects of PEI on memory are effective mainly in these subjects (e.g. RSA; Szpitalak & Polczyk, 2015).

From the perspective of eyewitness testimony practice, another aspect may also be important. Eyewitnesses may encounter several mutually contradictory pieces of information after an event. A new line of research on the influence of multiple pieces of mutually contradictory misinformation was opened by Blank's team (Blank et al., 2022). In a series of three experiments, they showed the complex effect of multiple pieces of misinformation. On one hand, becoming aware of internal contradictions between pieces of misinformation reduced the effects of these contradictions on memory. On the other hand, in the case of subtle misinformation, which is more difficult to capture, being exposed to multiple pieces of misinformation increases the chances that at least one of them will be noticed and then included in memory. It would be interesting to examine how multiple pieces of post-event information in the form of correct negation and misinformation affect witness memory. The present study suggests that explicit and implicit negation and misinformation have similar effects after a 1-week interval. It would therefore be worthwhile to test their combined effects on memory, taking into account longer and shorter time intervals and varying the distinctiveness of details (distinct misinformation is easier to detect, Polczyk, 2007).

Another interesting development in our research could be to add an additional phase to the procedure we use: participants' memory

would be tested before exposing them to post-event material. Extensive research on the phenomenon known as retrieval-enhanced suggestibility (RES; e.g. J. C. Chan et al., 2009; 2017), suggests that witnesses may be more susceptible to misinformation if their memory for the witnessed event was tested before they were exposed to the misinformation. It would be interesting to investigate whether, as in the case of misinformation, the initial testing enhances the NRFM effect.

Ethical standards

Declaration of conflicts of interest

Iwona Dudek has declared no conflicts of interest

Romuald Polczyk has declared no conflicts of interest

Ethical approval

All procedures performed in studies involving human participants were in accordance with the ethical standards of the Jagiellonian University Institute of Psychology Research Ethics Committee (opinion no. KE/35_2020) and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed consent

Informed consent was obtained from all individual participants included in the study

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